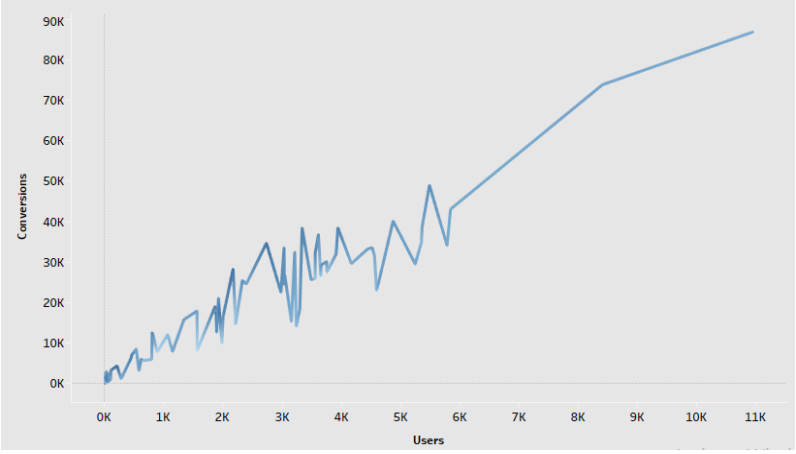
"Data Analysis and Insights for different page Optimization & How to get more user install & Engagement from the App & Website"

1. Traffic Acquisition Analysis :

Session default channel group	Conversions	Engaged sessions	Sessions	Users
Unassigned	114,161	1,481	13,448	20,263
Organic Search	33,612	17,814	21,241	7,689
Display	20,031	10,613	18,292	9,613
Direct	18,496	7,649	13,220	4,042
Paid Search	7,595	3,452	6,788	2,909
Organic Social	19	12	16	11

Viz.i : Scatter Plot of Users v/s Conversions



Positive Correlation Between Users and Conversions: The observed positive correlation between the number of users and conversions across these top categories is a promising trend. It implies that as user interactions increase, so does the likelihood of conversions.

Viz. ii : Dual axis chart for Users v/s engaged-session for each source of traffic acquisition

Delving into user engagement, this dual-axis chart unveils the dynamic interplay between user acquisition and engaged sessions across diverse traffic sources. Unveiling insights, it informs strategic decisions for optimized digital outreach.

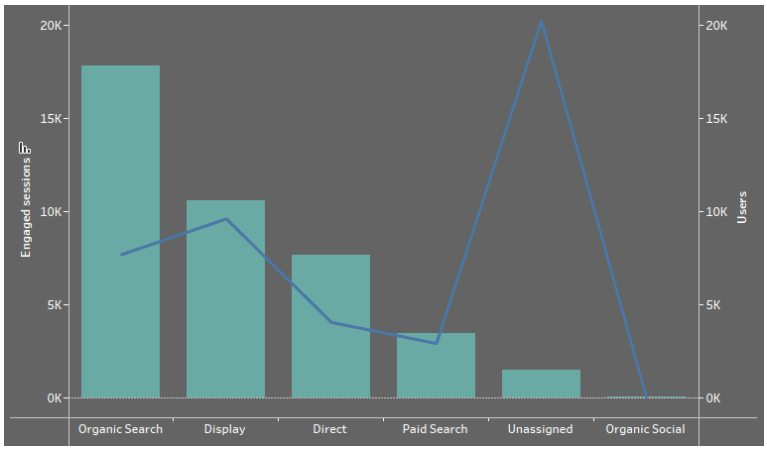


Fig : Here the line denotes “Users” and the bar chart “Engaged Sessions” for each category

INSIGHTS :

High Conversion Rate for Unassigned Sessions:
While Unassigned sessions have the highest number of users, their conversion rate is exceptionally high. This may indicate that a significant portion of users navigating without a defined source is still converting effectively.

Display and Organic Search Perform Well:
Display and Organic Search channels demonstrate good performance, with substantial user counts and respectable conversion rates. These channels contribute significantly to both user acquisition and conversions.

Direct Channel's Efficiency:
Direct sessions, while having a lower user count compared to Display and Organic Search, exhibit a commendable conversion rate. This suggests that users directly navigating to the platform have a higher likelihood of converting.

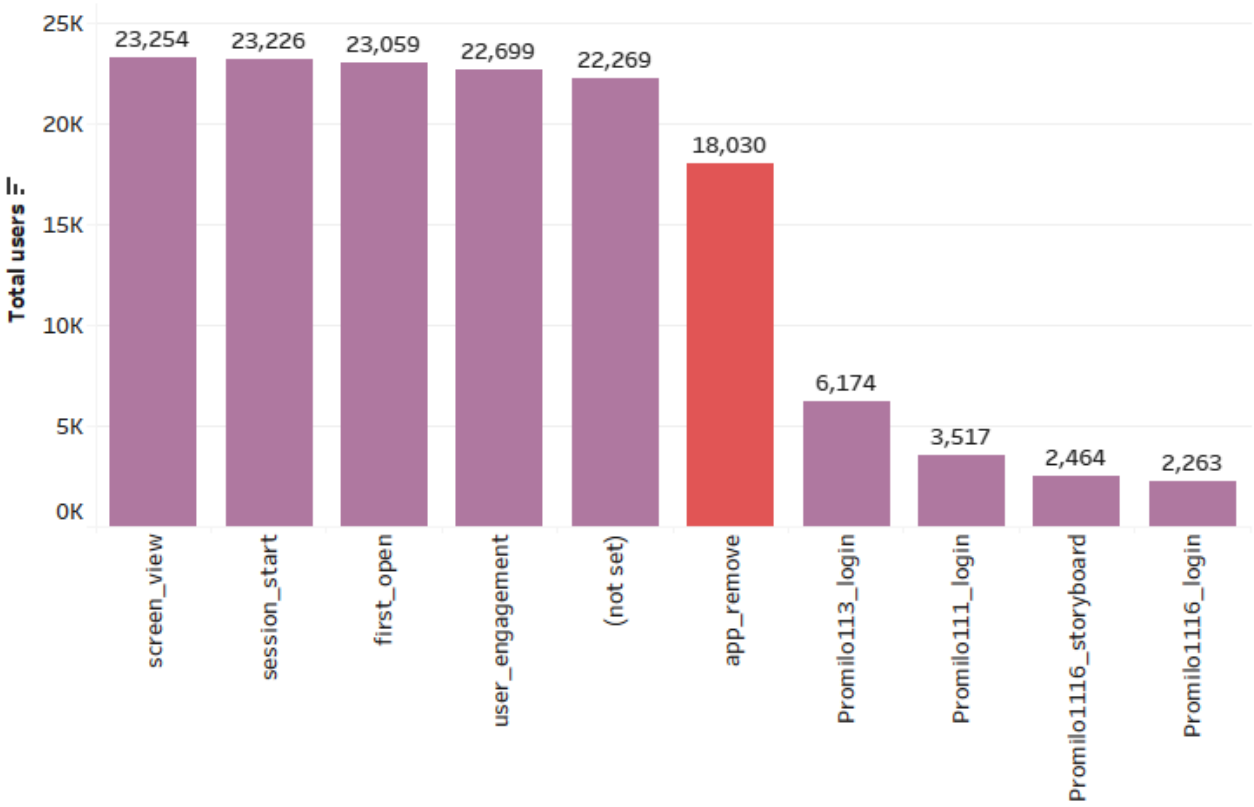
Paid Search's Contribution:
Paid Search, though with a relatively lower user count, contributes reasonably to conversions. This implies that the investment in paid advertising is translating into desired user actions.

Organic Social's Limited Impact:
Organic Social has a minimal presence, with low user numbers and conversions. This suggests that, at present, the organic social channel may not be a significant driver of traffic or conversions.

In summary, while Unassigned sessions surprisingly lead in conversions, Display, Organic Search, and Direct channels are noteworthy for their balanced performance in attracting users and converting them. Paid Search, despite a lower user count, contributes meaningfully to conversions, while Organic Social currently has limited impact. Considerations for optimizing strategies in each channel could further enhance overall performance.

2. Event Report Analysis:

Viz. :

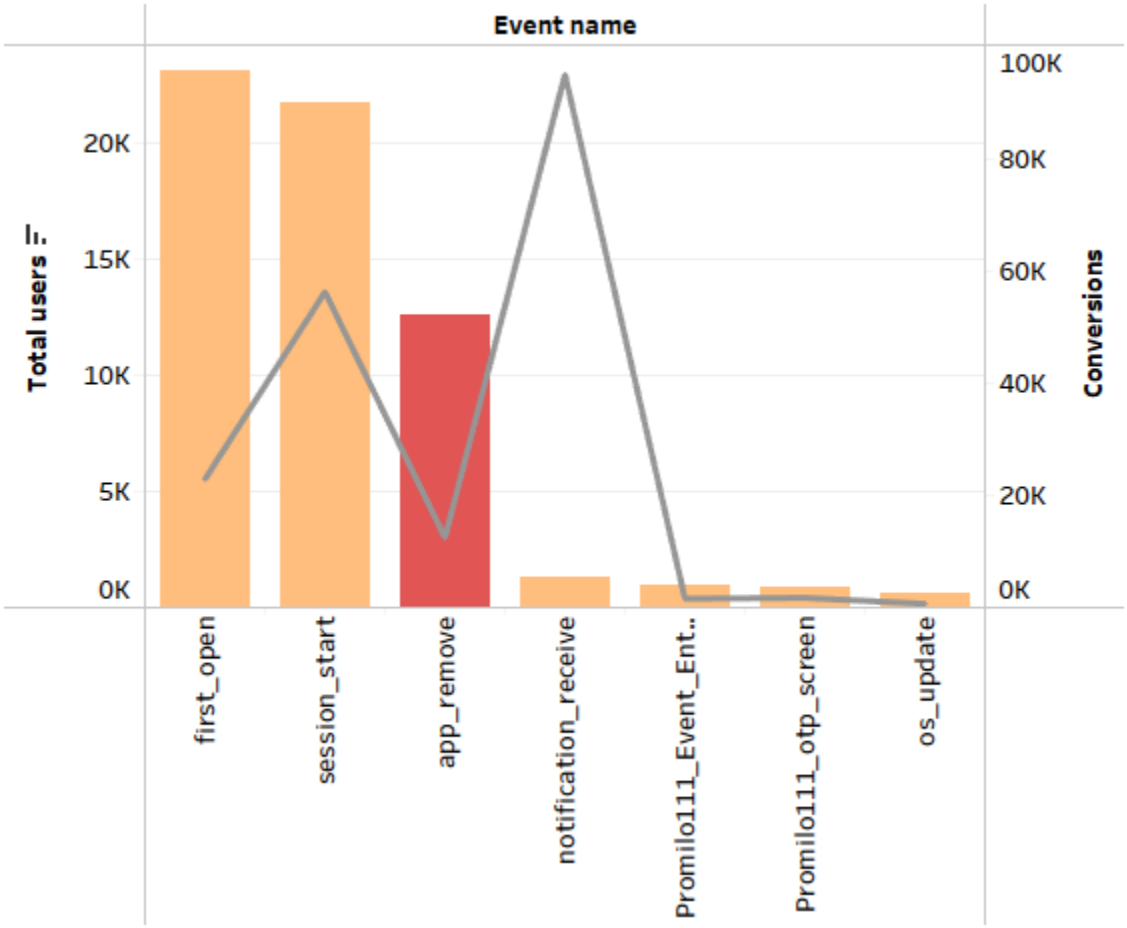


INSIGHTS :

The bar chart based on the Event Report table highlights notable patterns in user interactions. Key insights include the prevalence of "screen_view," "session_start," and "first_open," indicating user engagement. **However, the substantial count of 18,030 for "app_remove" raises concerns as it suggests a significant number of app uninstallations**. This demands in-depth analysis to understand the reasons behind such high uninstallation rates, potentially uncovering usability issues, unmet user expectations, or performance issues. **Addressing the root causes of app removal is crucial for enhancing user retention and optimizing the overall user experience**.

3. Conversion Report:

Viz. :



INSIGHTS :

Analyzing the conversion data for each event provides valuable insights into user engagement and conversion efficiency:

Notification_Receive - High Conversions, Low Users:
"notification_receive" stands out with a high conversion rate relative to the number of users, indicating effective engagement through notifications.

Session_Start - Significant Conversions and Users:
"session_start" shows a substantial number of both conversions and users, suggesting it is a critical point for user engagement, potentially signaling the start of meaningful interactions.

First_Open - High Conversion Rate, Close User Count:
"first_open" demonstrates a high conversion rate with a user count close to the number of conversions. This suggests that users who open the app for the first time are more likely to convert.

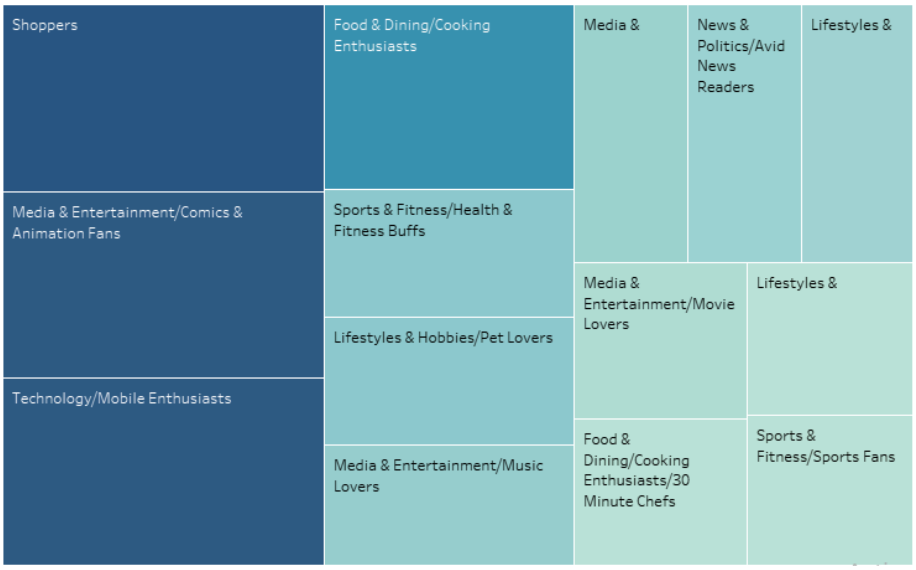
App_Remove - High Uninstallation Rate:

"app_remove" stands out with a high number of users and conversions, signifying app uninstalls. This event requires detailed analysis to identify reasons behind user attrition, potentially uncovering issues affecting user retention.

In summary, the analysis emphasizes the importance of understanding user behavior at different touchpoints, particularly during critical events like **app removal**. **Focusing on events with high conversion rates and addressing issues related to uninstallations can significantly improve overall user engagement and retention**.

4. User by Interest:

Viz. i : Top 15 categories which has more user interaction :



INSIGHTS :

Analyzing the top 15 categories with the highest user interactions reveals compelling insights:

Diverse User Interests:

The top 15 categories encompass a wide range of interests, from shopping ("Shoppers") and technology enthusiasts ("Technology/Mobile Enthusiasts") to media and entertainment enthusiasts with specific interests like comics, animation, music, and TV shows. This diversity suggests a broad user base with varied preferences.

Engagement Across Lifestyle and Hobbies:

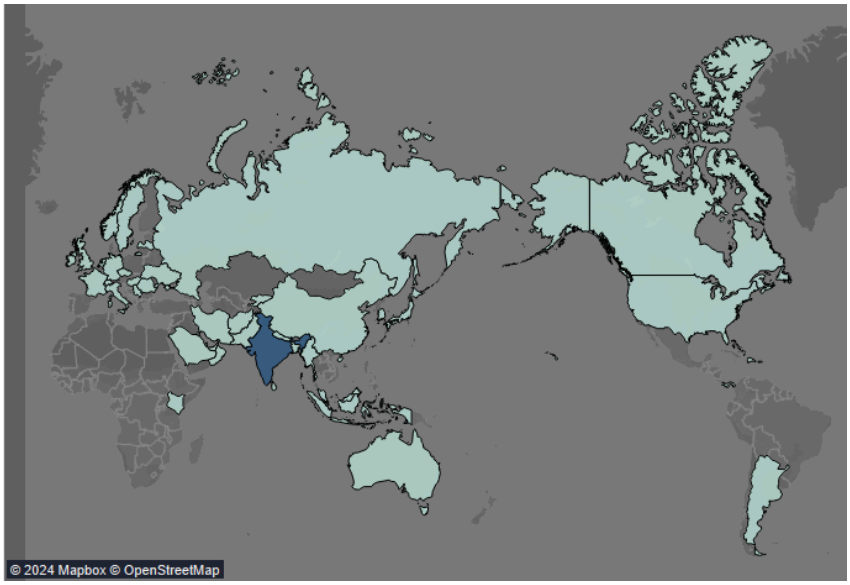
Categories such as "Lifestyles & Hobbies/Pet Lovers," "Lifestyles & Hobbies/Outdoor Enthusiasts," and "Sports & Fitness/Health & Fitness Buffs" highlight the platform's success in attracting users with lifestyle and wellness interests. This indicates a platform that resonates with users seeking content related to their hobbies and passions

Media & Entertainment Dominance:

Media and entertainment categories, including "Media & Entertainment/Comics & Animation Fans," "Media & Entertainment/Music Lovers," and "Media & Entertainment/TV Lovers," demonstrate a significant user engagement. This suggests a strong appeal for content related to comics, music, and television, showcasing the platform's entertainment value.

5. Demographic Analysis:

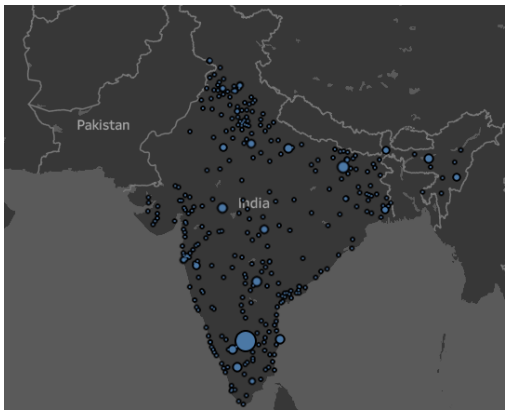
Viz. i :

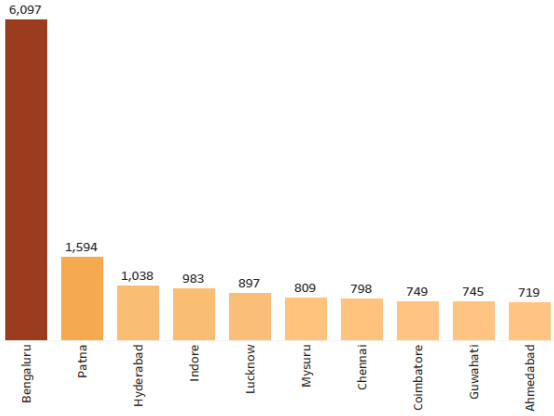


INSIGHTS :

The user distribution across countries reveals a substantial presence in **India, with 23,024 users, indicating a significant user base in this market**. While the United States follows with 272 users, suggesting a smaller but noteworthy presence. Canada, with 37 users, represents a more limited user base.

Viz. ii :



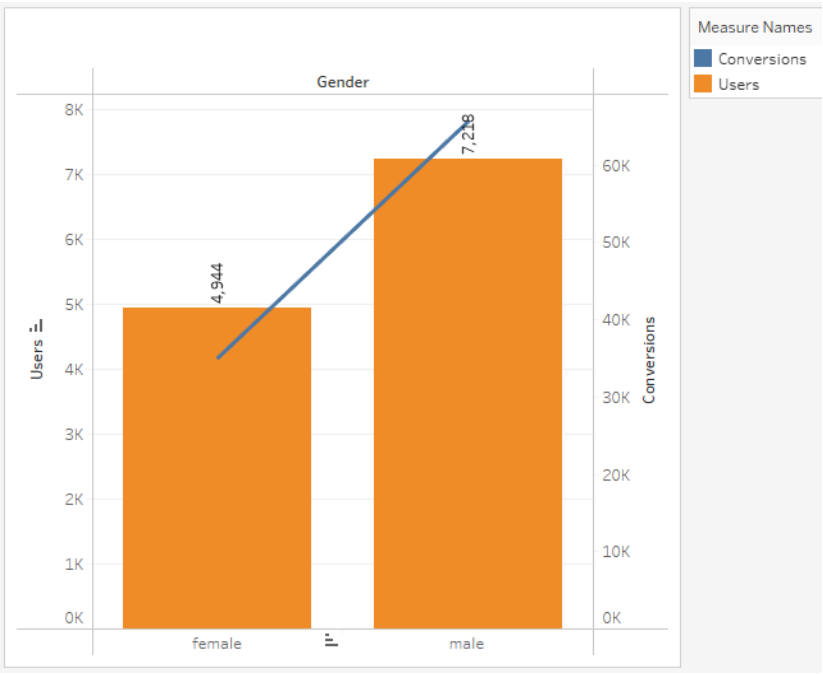


INSIGHTS :

The user distribution across various towns and cities reveals significant engagement in **Bengaluru, with 6,097 users,** indicating a substantial user base in the tech hub. **Patna follows with 1,594 users**, showcasing notable engagement in the capital of Bihar. Hyderabad and Indore have 1,038 and 983 users, respectively, indicating considerable presence.Other cities like Lucknow, Mysuru, Chennai, and Coimbatore also demonstrate noteworthy user engagement, highlighting diverse geographical reach and potential for tailored regional content strategies.

6. Gender-wise Analysis:

Viz. i :

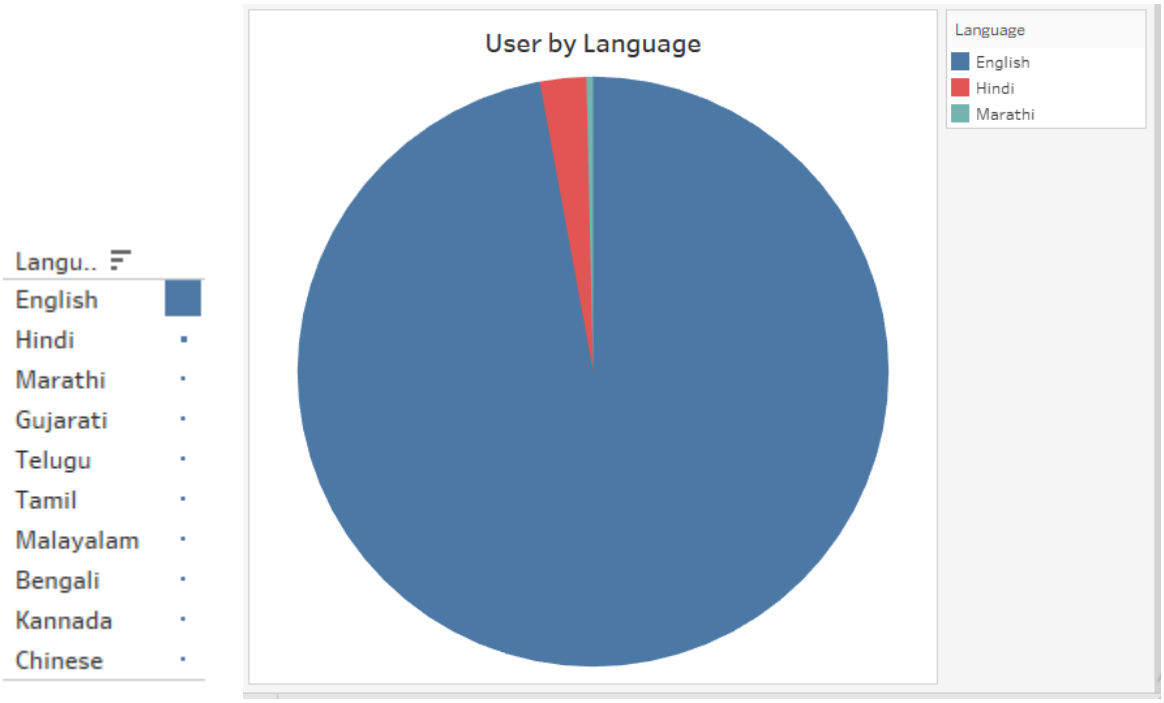


INSIGHTS :

Analyzing the user data by gender reveals interesting insights. While male users significantly outnumber female users, with 7,218 compared to 4,944, the conversion metrics present a noteworthy contrast. Despite their lower numbers, female users exhibit a higher conversion rate compared to their male counterparts. This suggests that the platform may have a more effective engagement strategy or content appeal for female users, leading to a relatively higher conversion rate. This insight underscores the importance of considering both user demographics and conversion metrics for targeted content creation and marketing strategies to optimize overall platform performance.

7. User by Language Analysis:

Viz. i :



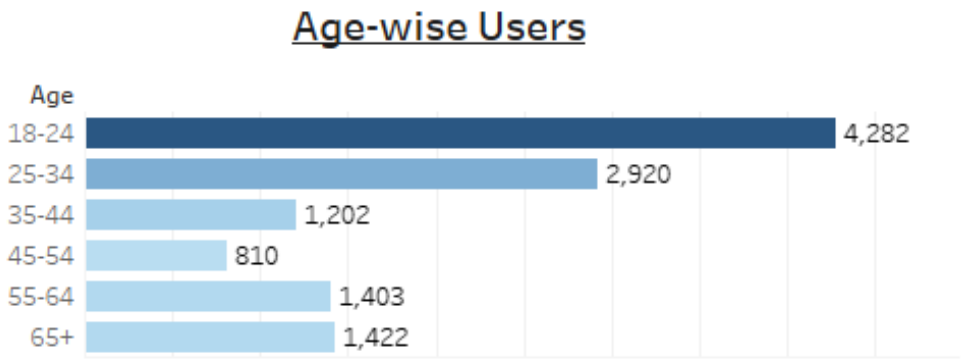
INSIGHTS :

The language distribution among users indicates a predominant preference for **English,** with a substantial user base of 22,495. **Hindi** follows with 586 users, reflecting a significant but comparatively smaller audience**. Marathi, Gujarati, Telugu, Tamil, Malayalam, Bengali, Kannada and Chinese** exhibit lower user counts, suggesting a diverse linguistic landscape. Understanding language preferences is crucial for content localization and engagement strategies, emphasizing the need for multilingual content to cater to the varied linguistic interests of the user base.

8. User by Age:

Viz. i :

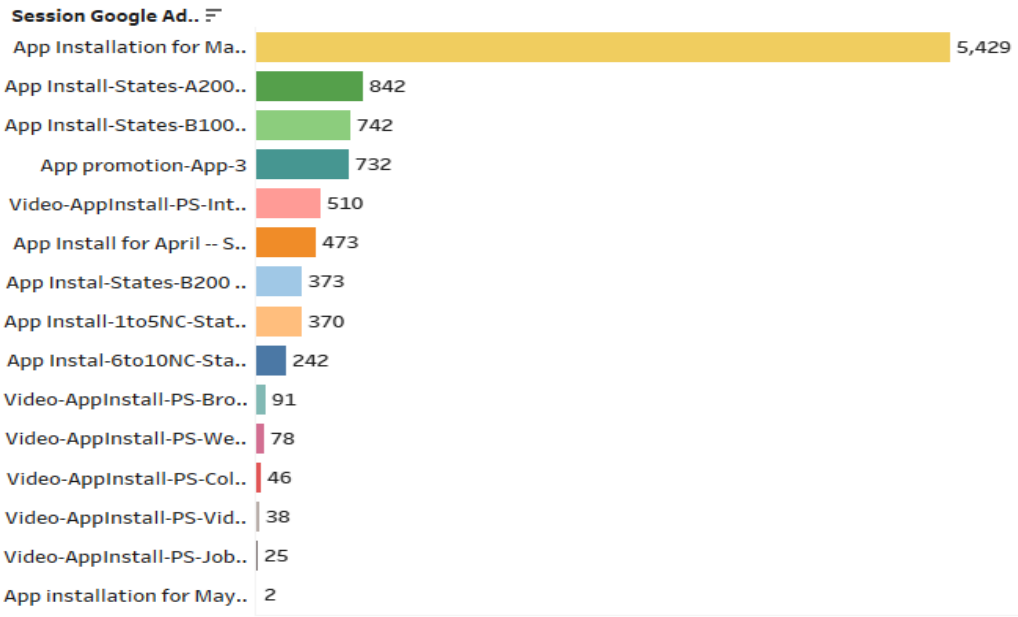
Age	Convers..	Users	Event count
18-24	53,661	4,282	309,328
25-34	20,172	2,920	90,074
35-44	8,111	1,202	33,016
45-54	2,946	810	18,661
55-64	4,823	1,403	25,169
65+	4,891	1,422	24,780



INSIGHTS : Analyzing user distribution by age reveals a diverse demographic on the platform. The age group **18-24** comprises the largest user segment with 4,282 users, reflecting a strong appeal to younger audiences. Users aged **25-34** also show substantial engagement, indicating a platform that resonates with young adults. Surprisingly, the **65+** age group demonstrates significant participation with 1,422 users, suggesting a broader age inclusivity. The platform's ability to attract a diverse age range highlights its broad appeal, emphasizing the need for varied content strategies to cater to the distinct preferences of different age groups.

9. Google Ads Analysis:

Viz. i :



INSIGHTS : Examining the Google Ads campaign performance reveals varying user responses across different app installation initiatives. The "App Installation for May --Shahid" campaign stands out with the highest user engagement at 5,429, suggesting its effectiveness. The campaigns ``App Install-States-A200 Inst-20Jun22" and "App Install-States-B100Installs-22Jun22" also exhibit substantial user interest with 842 and 742 users, respectively. However, campaigns such as "Video-AppInstall-PS-Internships-11Jul22" and "Video-AppInstall-PS-Colleges-11Jul22" have comparatively lower user engagement, indicating potential areas for improvement. Recommendations include allocating more resources to successful campaigns, refining targeting for underperforming ones, and experimenting with ad creatives and messaging to enhance overall campaign effectiveness.

SUMMARY OF INSIGHTS :

1. User-Conversions Correlation: Positive correlation observed; higher user interactions linked to increased conversion likelihood.

2. Channel Performance Summary: Unassigned leads in conversions; Display, Organic Search, and Direct channels balance attracting and converting users; Paid Search, despite lower users, contributes meaningfully; Organic Social has limited impact.

3. Event Report Insights: Bar chart highlights key user interaction patterns; "screen_view," "session_start," and "first_open" indicate robust engagement; concern raised by high "app_remove" count requires in-depth analysis for improved user retention.

4. Event-Specific Analysis:

Notification_Receive: High conversions, low users - effective engagement through notifications.

Session_Start: Significant conversions and users - critical point for meaningful interactions.

First_Open: High conversion rate, close user count - initial app opens correlate with higher conversion.

App_Remove: High uninstallation rate demands detailed analysis to address issues impacting user retention.

5. Diverse User Interests:

Top categories span shopping, technology, and media interests.

Indicates a broad user base with varied preferences.

6. Engagement Across Lifestyle and Hobbies:

Success in attracting users interested in lifestyle, wellness, and outdoor activities.

Platform resonates with users seeking content related to hobbies.

7. Media & Entertainment Dominance:

Significant engagement in categories like Comics & Animation, Music, and TV.

Highlights the platform's entertainment value.

8. Geographical Reach:

Strong presence in India (23,024 users) and notable engagement in cities like Bengaluru and Patna.

Diverse geographical reach suggests potential for tailored regional content strategies.

9. Gender Dynamics:

Male users outnumber females (7,218 to 4,944), but females exhibit a higher conversion rate.

Suggests a more effective engagement strategy or content appeal for female users.

10. Language Preferences:

English dominates with 22,495 users.

Emphasizes the need for multilingual content to cater to diverse linguistic interests.

11. Diverse Age Demographics:

Largest user segment in the 18-24 age group (4,282 users).

Surprising engagement from the 65+ age group (1,422 users) showcases broad age inclusivity.

12. Google Ads Campaign Performance:

"App Installation for May --Shahid" campaign stands out with 5,429 users.

Recommendations include resource allocation based on successful campaigns and refining targeting for underperforming ones.

RECOMMENDATION:

Based on the insights from the provided data, here are specific recommendations to boost app installations and website engagement:

1. Optimize Successful Google Ads Campaigns:

Allocate more resources to high-performing campaigns, such as "App Installation for May --Shahid." Increase ad spend and refine targeting to maximize user engagement and drive additional app installations.

2. Enhance Unassigned Channel Strategies:
Capitalize on the positive correlation between Unassigned channel interactions and conversions. Invest in optimizing strategies for the Unassigned channel to further increase conversions and user engagement.

3. Targeted Content for Top Categories:
Develop tailored content strategies for the top-performing categories, including "Shoppers," "Comics & Animation Fans," and "Mobile Enthusiasts." Create engaging and relevant content to attract more users and encourage app installations within these high-impact categories.

4. Address App Removal Concerns:
Investigate the high count of "app_remove" events to identify and address factors leading to app uninstalls. Implement improvements in user experience, address usability issues, and respond to user feedback to reduce app removal rates and enhance overall retention.

5. Regional Customization for Geographical Reach:
Leverage the significant user presence in India and engagement in cities like Bengaluru and Patna. Customize promotional efforts to align with regional preferences and interests. Tailor content and campaigns to resonate with users in different geographical locations.

6. Gender-Centric Marketing:
Design marketing initiatives targeting female users to capitalize on their higher conversion rates. Craft content and promotions that specifically appeal to female audiences, ensuring a more effective engagement strategy and potentially increasing app installations among this demographic.

7. Multilingual Content Development:
Expand language support and create content in multiple languages to cater to diverse linguistic preferences. This approach will broaden the app's appeal, attract a more diverse user base, and potentially lead to increased installations.

8. Diverse Age Demographic Strategies:
Tailor content and app features to meet the distinct preferences of different age groups. Develop strategies that resonate with the interests of users in the 18-24 age segment, as well as surprising engagement from the 65+ age group. This approach ensures a more inclusive user experience.
Implementing these targeted recommendations aligns with the specific insights derived from the data and aims to optimize user engagement and drive more app installations based on observed trends and correlations. Regularly monitor performance metrics and user feedback to iteratively refine strategies for sustained improvement.

TOOLS USED :

Python : Performing EDA and cleaning data
Google Bigquery : Extracting important insights in SQL
Tableau : Visualization